

Cities within corrected 100 km member reach

Analysis map 12 documentation

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Science summary

This map describes geographic reach of corrected member buffers to mapped urban centres. It identifies nearby city context and potential partners. A 100 km buffer does not establish travel time, accessibility, service delivery or ecological impact. The saved geodesic buffers are reused with their geometry preserved.

Inputs and calculation

Intersect Members100km with Cities and count distinct nonblank registration IDs for each city. Count each city once per buffer summary. The city reach fields are zero for unreached cities, rather than missing. Whole-city population uses Climate / GC_POP_TOT_2025; this differs from the infrastructure population GH_POP_TOT_2025 used by maps 01–11.

Field	Definition
GS_Reach100N	Distinct registration IDs with intersecting 100 km buffers
GS_Reach100CityN	Distinct reached cities
GS_Reach100Pop	Sum of whole-city GC_POP_TOT_2025 populations
GS_Reach100Green / Heat / Flood	Equal-city means of valid GreenPop, HeatDays and FloodPop
GS_Reach100Rank / Pctile / RankN	Descending count rank, midpoint percentile and valid N
GS_Reach100Short / Long	Concise popup and longer sidebar summary

Jurisdiction attribution

Reach summaries at diocesan and conference scale retain the prior unique-city assignment method. A city is included only if it intersects exactly one jurisdiction at the relevant scale. This is intentionally different from maps 01–11, where all distinct intersecting cities contribute to means. Boundary-crossing and unassigned cities are reported as exclusions; no nearest-jurisdiction assignment is substituted.

Ecclesiastical information

All 43 diocesan and 186 conference original attributes were preserved and compared value-for-value. Names, rites, traditions, hierarchy, source dates, population, Catholic statistics, clergy, parishes, leadership, enrolment, education and duplicate source series remain available. Member and buffer source attributes and geometry also remain. GS_Ecclesial provides readable context in longer summaries. Leadership and historic statistics retain source dates; they were not revalidated as current officeholders. Fields with suffix _26 and other original series were not merged or silently overwritten.

Results and exclusions

Measure	Result
Buffer-city relationships	26,929
100 km buffers containing a city	2,666 of 2,739
Buffers without city intersections	73
Distinct reached cities	4,027
Diocesan uniquely assigned reached cities	3,547; 470 cross-diocese and 10 unassigned excluded
Conference uniquely assigned reached cities	3,957; 60 cross-conference and 10 unassigned excluded

Rankings and display

Rank 1 is the highest numeric value, with competition ranks for ties. Percentile is $100 \times (\text{number below} + \text{half the number equal}) / \text{valid N}$. Higher percentile means a higher numeric value, not better or worse conditions. Missing values are excluded. All-equal values have rank 1 and percentile 50.

All 11,422 cities, including zero-reach cities, form the city comparison group. Dioceses and conferences are separately ranked by their own saved reach counts. Graduated circles use 4,027 city centroid display features and five quantile size classes; circle size is not a continuous proportional area scale. Original city polygons remain available.

Quality control

A new read-only check reproduced nine saved reach fields for every city, diocese and conference using explicit saved relationships and final exports. All 11,422 city records, 2,874 dioceses and 122 conferences matched with zero differences. Counts, whole-city population totals, environmental means, competition ranks, percentiles and comparison-group sizes passed. Evidence:

../13_Source_Data_Documentation/Evidence/reach_documentation_qc.json. This check does not independently certify buffer geometry or the publisher's population model.

Project checks: no broken layers; saved reach popup fields verified. Existing reach_qc.json records exclusions and totals. Geometry preservation checks support reuse of the CorrectGeo buffers.

Reproduction and limitations

Open ../LSAP_Global_Human_Settlements.aprx and map 12. Use archived calculate_reach.py, saved relationships and the source inventory. Whole city populations are included even when only part of a city intersects a buffer; they are not populations strictly within 100 km. Overlapping buffers can reach the same people and places, so buffer totals are not additive. Registration IDs are distinct nonblank IDs rather than point-record counts. GS_Reach100Long supplies sidebar text, and original jurisdiction fields provide the ecclesiastical context.